



# s2xout:1.14

API Reference

X Platform, S2X Out 2.14.0

05/05/2025

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# 1 Overview

## Changelog

### 1.14

#### Changed

- s2xoutStatus from 1.2 to 1.3

### 1.13

#### Changed

- outputMapping from 1.4 to 1.5
- s2xoutModulator from 1.12 to 1.13

### 1.11

#### Changed

- tsstatus from 1.3 to 1.5 to have access to the rpc GetOutputPidStatus.
- component from 1.4 to 1.5
- genericDescriptor from 1.1 to 1.2

#### Removed

- The external dependency status, due to it not being needed anymore on this module.

### 1.10

#### Added

- satellite mode of profile configuration (see **Satellite mode** section below)

#### Changed

- s2xoutPortProfile from 1.0 to 1.1
- s2xoutPhysicalPort from 1.2 to 1.3
- s2xoutModulator from 1.9 to 1.10

### Satellite mode

Until version 1.10 there was a single way of configuring port profiles where user has direct control over parameters of the output signal (“what you set it what you get”).

Version 1.10 of the API introduced an alternative way of configuring carrier frequency which is intended to make it easier to configure satellite uplink.

Instead of user needing to manually calculate appropriate carrier frequency based on the desired satellite transmission frequency and frequency of Block Upconverter (BUC) device, now it's possible to configure **satellite transmission frequency** and provide **BUC frequency** of external Block Upconverter device and the system will automatically calculate proper carrier frequency for the output signal (and also automatically determine spectrum inversion).

Satellite mode of configuration is optional. The direct mode where user directly configures carrier frequency and spectrum inversion is still in place.

### How to use satellite mode of configuration?

- user needs to set BUC frequency - bucFrequency field in PhysicalPort type (see *Type Reference* in *s2xoutPhysicalPort* section)
- port profiles are now divided into two categories - direct and satellite, the distinction is based

on RfSettings variant

- see RfSettings, RfSettingsDirect, RfSettingsSatellite in *Type Reference of s2xoutPortProfile*
- direct profiles can be used as before but only if BUC frequency is NOT set on a physical port
- satellite profiles require BUC frequency to be set on a physical port
- if frequency override is configured on a physical port (to override frequency of a profile) and if a port uses satellite profile then the override will affect satellite frequency, so in such case it's effectively satellite frequency override (see frequency field of RfOverrides in *s2xoutPhysicalPort* section)

#### Useful details about satellite mode

- output mode is automatically set to L-BAND - this means that valid range of satellite frequencies will be: BUC frequency +/- L-BAND range
- user does not configure inverted spectrum parameter and it will be automatically determined by the system based on relation between satellite transmission frequency and BUC frequency (if satellite frequency is smaller than BUC frequency then spectrum will be inverted)

## 2 bissCaSettings (1.0)

### 2.1 Type Reference

#### 2.1.1 BissCaEcmInterval

BISS CA ECM Repetition Interval

Predefined set of possible ECM repetition interval values, in milliseconds.

Values lower than 100 ms are outside of the BISS CA specification. These values may still be used if the parameter `extendedOptions` is set to true in the `BissCaSettings` struct.

##### enum

|      |                                      |
|------|--------------------------------------|
| I6   | 5.7 ms (Value outside BISS CA spec)  |
| I15  | 14.7 ms (Value outside BISS CA spec) |
| I50  | 50 ms (Value outside BISS CA spec)   |
| I100 | 100 ms                               |
| I125 | 125 ms                               |
| I150 | 150 ms                               |
| I175 | 175 ms                               |
| I200 | 200 ms                               |
| I250 | 250 ms                               |
| I300 | 300 ms                               |

#### 2.1.2 BissCaEntitlementFlags

BISS CA Entitlement Flags

The entitlement flags are parameters for how the entitled receivers shall behave. These values are mandatory part of the stream, and is inserted into the session data, which is the encrypted part of the EMM table.

##### struct

|                                        |                                             |
|----------------------------------------|---------------------------------------------|
| <code>preventDescrambledForward</code> | <b>bool</b><br>Prevent descrambling forward |
| <code>preventDecodedForward</code>     | <b>bool</b><br>Prevent decode forward       |
| <code>insertWatermark</code>           | <b>bool</b><br>Insert watermark             |

#### 2.1.3 BissCaSettings

BISS CA Specific Settings

##### struct

|                           |                                           |
|---------------------------|-------------------------------------------|
| <code>cryptoPeriod</code> | <b>int</b><br>Crypto period (in seconds). |
| <code>emmBitrate</code>   | <b>int</b><br>EMM bitrate (kbps).         |
|                           | <b>BissCaEcmInterval</b>                  |

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ecmInterval      | ECM repetition interval. Note that changes to the repetition interval will not take effect until the next crypto period (the next time the ECM content is updated).                                                                                                                                                                                                                                                  |
| emmPreferredPid  | <b>int</b><br>EMM preferred PID value. If the PID value is not available, the first larger PID value available will be used instead. If no larger PID value is available, then the first smaller available value will be used.                                                                                                                                                                                       |
| ecmPreferredPid  | <b>int</b><br>ECM preferred PID value. If the PID value is not available, the first larger PID value available will be used instead. If no larger PID value is available, then the first smaller available value will be used.                                                                                                                                                                                       |
| entitlementFlags | <b>BissCaEntitlementFlags</b><br>BISS CA entitlement flags. Controls how the entitled receivers shall behave.                                                                                                                                                                                                                                                                                                        |
| extendedOptions  | <b>bool</b><br>Allow for extended options. This means allowing violation of the BISS CA specification. If set to true, then the following requirements of the BISS CA specification are <i>not</i> enforced: - EMM repetition interval must be at least 200 ms - ECM repetition interval must be at least 100 ms Note that the requirement that EMM interval must be larger than the ECM interval is still enforced. |

### 3 bissHelpers (1.0)

## 4 bissScrambling (1.6)

### 4.1 Type Reference

#### 4.1.1 ScramblerSettings

- Scrambler Settings
- 
- Settings defining how to scramble the transport stream.
- By default, there is a single scrambling settings parameter, where these
- settings are applied to all services in the transport stream (TS).
- 
- Optionally, however, one or more extra settings may be configured, where the
- different settings are applied to different services in the TS. The different
- settings are referenced by integer indecies. These indecies are
- cross-references to the transport stream configuration. For example, a single
- service may refer to one of these indecies, meaning that the service will be
- scrambled with the corresponding scrambling settings.
- 
- Requirements:
  - – All extra scrambling setting *must* contain the same scrambling method as
- the main settings.
- 
- If these requirements are not met, the configuration is not valid.
- 
- 

#### struct

|         |                                                                                                                   |
|---------|-------------------------------------------------------------------------------------------------------------------|
| enabled | <b>bool</b><br>Scrambling on/off<br>•                                                                             |
| main    | <b>ScramblingMethod</b><br>The main (default) scrambling settings<br>•                                            |
| extra   | <b>map</b> from <b>int</b> to <b>ScramblingMethod</b><br>Additional scrambling settings, each with its own index. |

#### 4.1.2 ScramblingMethod

ScramblingMethod

**variant**

|             |                                                                |
|-------------|----------------------------------------------------------------|
| fixedKeyRaw | <b>bissTypes.FixedKeyRawScrambling</b><br>Fixed key encryption |
| biss1       | <b>bissVersion1.Scrambling</b><br>BISS version 1 encryption    |
| biss2       | <b>bissVersion2.Scrambling</b><br>BISS version 2 encryption    |

## 5 bissSettings (1.0)

### 5.1 Type Reference

#### 5.1.1 AllowedExtraSettings

Allowed Num of Extra Scrambling Settings

##### enum

|               |                   |
|---------------|-------------------|
| EXTRA_15_ECMS | 15 extra settings |
| EXTRA_7_ECMS  | 7 extra settings  |
| EXTRA_3_ECMS  | 3 extra settings  |

#### 5.1.2 AllowedMode

Allowed (De)Scrambling modes

##### enum

|          |                                |
|----------|--------------------------------|
| ALL      | CSA and AES algorithms allowed |
| AES_ONLY | Only AES allowed               |
| CSA_ONLY | Only CSA allowed               |

#### 5.1.3 BissModuleSettings

BISS Module Setup for a given product (not configurable)

##### struct

|                      |                                                                                                                                    |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------|
| allowedMode          | <b>AllowedMode</b><br>Algorithms supported by product                                                                              |
| numExtraPerOutput    | <b>AllowedExtraSettings</b><br>Number of extra settings allowed per output                                                         |
| maxNumOutputs        | <b>int</b><br>Maximum number of outputs                                                                                            |
| totalEmmBitrateLimit | <b>optional int</b><br>Set an upper limit for the total EMM playout bitrate (sum of the bitrate all EMMs). Given in units of kbps. |

## 6 bisstypes (1.3)

### 6.1 Type Reference

#### 6.1.1 Algorithm

De/Scrambling Algorithm

The DVB-CSA algorithm uses a 64-bit key, while all the AES algorithms use a 128-bit key.

##### enum

|                 |                                            |
|-----------------|--------------------------------------------|
| DVB_CSA         | DVB-CSA                                    |
| AES_ATIS_IDSA   | AES ATIS IDSA                              |
| AES_ECBT_ECB1   | AES-ECBT/AES-ECB1 (Alticast)               |
| AES_ECBL_ECB2   | AES-ECBL/AES-ECB2 (Latens)                 |
| AES_CBCT_CBC1   | AES CBCT CBC1                              |
| AES_DVB_CISSA   | AES DVB-CISSA                              |
| AES_IRDETO_CBCT | AES CBCT Irdeto                            |
| AES_CBCT_IV     | AES CBCT w/specified Initialization Vector |

#### 6.1.2 FixedKeyRawDescrambling

Descrambling Method: Fixed Key Raw

##### struct

|                 |                                                                                   |
|-----------------|-----------------------------------------------------------------------------------|
| algorithm       | <b>Algorithm</b><br>Scrambling algorithm                                          |
| key             | <b>string</b><br>Scrambling key                                                   |
| iv              | <b>optional string</b><br>Scrambling Initialization vector (only for AES_CBCT_IV) |
| serviceMappings | <b>list of FixedKeyServiceMapping</b><br>A list of service mappings.              |

#### 6.1.3 FixedKeyRawScrambling

Scrambling Method: Fixed Key Raw

##### struct

|           |                                                                                   |
|-----------|-----------------------------------------------------------------------------------|
| algorithm | <b>Algorithm</b><br>Scrambling algorithm                                          |
| key       | <b>string</b><br>Scrambling key                                                   |
| iv        | <b>optional string</b><br>Scrambling Initialization vector (only for AES_CBCT_IV) |

#### 6.1.4 FixedKeyServiceMapping

Fixed key descrambling for a list of services

**struct**

|            |                                                                                   |
|------------|-----------------------------------------------------------------------------------|
| key        | <b>string</b><br>Scrambling key                                                   |
| iv         | <b>optional string</b><br>Scrambling Initialization vector (only for AES_CBCT_IV) |
| serviceIds | <b>list of int</b><br>A list of services that the key applies to.                 |

## 7 bissValidate (1.0)

### 7.1 Type Reference

#### 7.1.1 FloatRange

Float Range

**struct**

|     |                                            |
|-----|--------------------------------------------|
| min | <b>float</b><br>Minimum value of the range |
| max | <b>float</b><br>Maximum value of the range |

#### 7.1.2 IntRange

Integer Range

**struct**

|     |                                          |
|-----|------------------------------------------|
| min | <b>int</b><br>Minimum value of the range |
| max | <b>int</b><br>Maximum value of the range |

#### 7.1.3 Result

Validation result

**struct**

|        |                                                                                          |
|--------|------------------------------------------------------------------------------------------|
| result | <b>bool</b><br>True if config is valid and false otherwise.                              |
| msg    | <b>optional string</b><br>An error message, which may be given if the result is invalid. |

## 8 bissVersion1 (1.3)

### 8.1 Type Reference

#### 8.1.1 CsaKeyExpansion

Method of CSA Key Expansion from 48 to 64 bit

##### enum

|          |                                          |
|----------|------------------------------------------|
| CHECKSUM | Session Word is expanded using checksum. |
| ZERO     | Session Word is expanded using zeros.    |

#### 8.1.2 Descrambling

BISS version 1 Descrambling

##### variant

|       |                                                     |
|-------|-----------------------------------------------------|
| mode1 | <b>ModelDescrambling</b><br>BISS version 1, Mode 1. |
| modeE | <b>ModeEDescrambling</b><br>BISS version 1, Mode E. |

#### 8.1.3 ModelDescrambling

BISS version 1 Descrambling, Mode 1

##### struct

|                 |                                                                          |
|-----------------|--------------------------------------------------------------------------|
| expansion       | <b>CsaKeyExpansion</b><br>Expansion method.                              |
| sessionWord     | <b>string</b><br>Session Word, 48-bit hex string.                        |
| serviceMappings | <b>list</b> of <b>ModelServiceMapping</b><br>A list of service mappings. |

#### 8.1.4 ModelScrambling

BISS version 1 Scrambling, Mode 1

##### struct

|             |                                                   |
|-------------|---------------------------------------------------|
| expansion   | <b>CsaKeyExpansion</b><br>Expansion method.       |
| sessionWord | <b>string</b><br>Session Word, 48-bit hex string. |

#### 8.1.5 ModelServiceMapping

BISS version 1 Descrambling, Mode 1 for a list of services

**struct**

|             |                                                                            |
|-------------|----------------------------------------------------------------------------|
| sessionWord | <b>string</b><br>Session Word, 48-bit hex string.                          |
| serviceIds  | <b>list of int</b><br>A list of services that the session word applies to. |

**8.1.6 ModeEDescrambling**

BISS version 1 Descrambling, Mode E

**struct**

|                 |                                                                             |
|-----------------|-----------------------------------------------------------------------------|
| activeId        | <b>UUID</b><br>Active session ID.                                           |
| expansion       | <b>CsaKeyExpansion</b><br>Expansion method.                                 |
| encSessionWord  | <b>UUID</b><br>Encrypted Session Word, 128-bit. Reference to BissData UUID. |
| serviceMappings | <b>list of ModeEServiceMapping</b><br>A list of service mappings.           |

**8.1.7 ModeEScrambling**

BISS version 1 Scrambling, Mode E

**struct**

|             |                                                                   |
|-------------|-------------------------------------------------------------------|
| expansion   | <b>CsaKeyExpansion</b><br>Expansion method.                       |
| sessionWord | <b>UUID</b><br>Session Word, 128-bit. Reference to BissData UUID. |

**8.1.8 ModeEServiceMapping**

BISS version 1 Descrambling, Mode E for a list of services

**struct**

|                |                                                                                                     |
|----------------|-----------------------------------------------------------------------------------------------------|
| activeId       | <b>UUID</b><br>Active session ID. Reference to BissData UUID.                                       |
| encSessionWord | <b>UUID</b><br>Encrypted Session Word, 128-bit. Reference to BissData UUID.                         |
| serviceIds     | <b>list of int</b><br>A list of services that the session ID and encrypted session word applies to. |

**8.1.9 Scrambling**

BISS version 1 Scrambling

**variant**

|       |                                                   |
|-------|---------------------------------------------------|
| mode1 | <b>Mode1Scrambling</b><br>BISS version 1, Mode 1. |
| modeE | <b>ModeEScrambling</b><br>BISS version 1, Mode E. |

## 9 bissVersion2 (1.6)

### 9.1 Type Reference

#### 9.1.1 Descrambling

BISS version 2 Descrambling

##### variant

|        |                                                       |
|--------|-------------------------------------------------------|
| mode1  | <b>ModelDescrambling</b><br>BISS version 2, Mode 1.   |
| modeE  | <b>ModeEDescrambling</b><br>BISS version 2, Mode E.   |
| modeCA | <b>ModeCADescrambling</b><br>BISS version 2, Mode CA. |

#### 9.1.2 ModelDescrambling

BISS version 2 Descrambling, Mode 1

##### struct

|                 |                                                                          |
|-----------------|--------------------------------------------------------------------------|
| sessionWord     | <b>string</b><br>Session Word, 128-bit hex string.                       |
| serviceMappings | <b>list</b> of <b>ModelServiceMapping</b><br>A list of service mappings. |

#### 9.1.3 ModelScrambling

BISS version 2 Scrambling, Mode 1

##### struct

|             |                                                    |
|-------------|----------------------------------------------------|
| sessionWord | <b>string</b><br>Session Word, 128-bit hex string. |
|-------------|----------------------------------------------------|

#### 9.1.4 ModelServiceMapping

BISS version 2 Descrambling, Mode 1 for a list of services

##### struct

|             |                                                                                   |
|-------------|-----------------------------------------------------------------------------------|
| sessionWord | <b>string</b><br>Session Word, 128-bit hex string.                                |
| serviceIds  | <b>list</b> of <b>int</b><br>A list of services that the session word applies to. |

#### 9.1.5 ModeCADescrambling

BISS version 2 Descrambling, Mode CA

**struct**

key

**optional UUID**

Private RSA key. If UUID value is given, this references a Biss-Data UUID. If optional is empty, the buried RSA private key is used.

**9.1.6 ModeCAscrambling**

BISS version 2 Scrambling, Mode CA

**struct**

receivers

**UUID**

Public RSA keys of receivers. Reference to BissData CSV File UUID or BissDataGroup UUID.

esId

**string**

Entitlement session ID (16-bit hex string).

settings

**bissCaSettings.BissCaSettings**

ModeCa settings.

**9.1.7 ModeEDescrambling**

BISS version 2 Descrambling, Mode E

**struct**

activeId

**UUID**

Active session ID. Reference to BissData UUID.

encSessionWord

**UUID**

Encrypted Session Word, 128-bit. Reference to BissData UUID.

serviceMappings

**list of ModeEServiceMapping**

A list of service mappings.

**9.1.8 ModeEScrambling**

BISS version 2 Scrambling, Mode E

**struct**

sessionWord

**UUID**

Session Word, 128-bit. Reference to BissData UUID.

**9.1.9 ModeEServiceMapping**

BISS version 2 Descrambling, Mode E for a list of services

**struct**

activeId

**UUID**

Active session ID. Reference to BissData UUID.

encSessionWord

**UUID**

Encrypted Session Word, 128-bit. Reference to BissData UUID.

serviceIds

**list of int**

A list of services that the session ID and encrypted session word applies to.

### 9.1.10 Scrambling

BISS version 2 Scrambling

**variant**

mode1

**Mode1Scrambling**

BISS version 2, Mode 1.

modeE

**ModeEScrambling**

BISS version 2, Mode E.

modeCA

**ModeCAscrambling**

BISS version 2, Mode CA.

## 10 component (1.5)

### 10.1 Overview

Changelog:

1.5

- Added
  - SCTE27\_SUB to ComponentType enum and respective mappings.

### 10.2 Type Reference

#### 10.2.1 AnyComponent

empty **struct**

#### 10.2.2 ComponentGroup

ComponentGroup: Grouping of ComponentType, used for filtering

**enum**

UNDEFINED  
AUDIO  
VIDEO  
SUBTITLE  
DATA

#### 10.2.3 ComponentLanguage

**string**

#### 10.2.4 ComponentSelector

Component selector.

|                 |                                                           |
|-----------------|-----------------------------------------------------------|
| @param pid      | Selects a component using a specific (source) PID.        |
| @param language | Selects a component by a three character language string. |
| @param type     | Selects a component by its type.                          |
| @param group    | Selects a component by group membership.                  |
| @param any      | Selects any component.                                    |

**variant**

|                |                          |
|----------------|--------------------------|
| pid            | <b>Pid</b>               |
| language       | <b>ComponentLanguage</b> |
| componentType  | <b>ComponentType</b>     |
| componentGroup | <b>ComponentGroup</b>    |
| any            | <b>AnyComponent</b>      |

### 10.2.5 ComponentType

A ComponentType is typically identified by a stream type (see ISO 13818-1 Table 2-29), often distinguished with another header field such as a descriptor tag.

#### enum

|                |
|----------------|
| MPEG_VIDEO     |
| MPEG_AUDIO     |
| TTX            |
| DVBSUB         |
| ECM            |
| PCR            |
| PRIVATE        |
| AC_3           |
| H264           |
| VBI            |
| AAC            |
| VC_1           |
| SCS_EMM        |
| DPI_CUE        |
| EAC3           |
| AIT            |
| HBBTV_CAROUSEL |
| DATA_CAROUSEL  |
| H265           |
| S302M          |
| JPEG2000       |
| S2038          |
| DVBTTMLSUB     |
| JPEG_XS_VIDEO  |
| SCTE27_SUB     |

### 10.2.6 Pid

#### int

### 10.2.7 SubtitleFormat

#### enum

|                     |
|---------------------|
| DVB                 |
| DVB_HARD_OF_HEARING |
| EBU                 |
| EBU_HARD_OF_HEARING |

## 11 componentMapping (1.3)

### 11.1 Overview

Changelog:

#### 1.3

- **Changed**
- component bumped from 1.4 to 1.5

#### 1.2

- **Changed**
- Moved the StreamType definitions to dedicated IDCL module, streamtype
- Now using the ComponentType definition found in IDCL module component, as the one found in componentMapping version 1.1 was redundant.

### 11.2 Type Reference

#### 11.2.1 ComponentAction

**enum**

STOP

PASSTHROUGH

REMAP

#### 11.2.2 ComponentGeneration

A pmt entry generation rule. Usually used to indicate that some external PID will Appear at the specified PID.

**struct**

|            |                                                                           |
|------------|---------------------------------------------------------------------------|
| streamType | <b>streamtype.StreamType</b><br>Streamtype as specified in the standards. |
| pid        | <b>int</b><br>PID number to signal the component on.                      |

#### 11.2.3 ComponentLayoutMap

A collection of a priority list of component mapping rules, and one default action to handle unmatched components.

**struct**

|               |                                                                                                                           |
|---------------|---------------------------------------------------------------------------------------------------------------------------|
| defaultAction | <b>ComponentMapDefaultAction</b><br>Default action to perform on any unmatched component                                  |
| layout        | <b>list</b> of <b>ComponentMapping</b><br>Priority list of mapping rules - each rule matches exactly one source component |

## 11.2.4 ComponentMapDefaultAction

### enum

|             |
|-------------|
| PASSTHROUGH |
| STOP        |

## 11.2.5 ComponentMapping

The ComponentMapping object describes a rule that will match one - and only one - incoming component, and perform an action on it. I.e, the selection of type MPEG\_AUDIO with language "nor" with an action set to "STOP", will stop one "instance" of mpeg audio with the norwegian language. Subsequent components matching the same filter will fall through to the next lower priority rule matching them.

There are some extra considerations in regards to pids containing PCR. It is not allowed to STOP the PCR content. If such a rule is applied, the payload of the component will be stopped, but the PCR content will still be mapped out.

It is possible to remap only the PCR content. To do this, componentType must be specified as PCR. Any other rule matching the same pid will remap all the content, not just the PCR.

### struct

|               |                                                                          |
|---------------|--------------------------------------------------------------------------|
| componentType | <b>optional component.ComponentType</b><br>Component type selector.      |
| language      | <b>optional string</b><br>Language selector.                             |
| pid           | <b>optional int</b><br>Pid selector.                                     |
| action        | <b>ComponentAction</b><br>What to do with the matched component.         |
| targetPid     | <b>optional int</b><br>Optional. Must be specified when action is REMAP. |

## 12 firewallTypes (1.0)

### 12.1 Type Reference

#### 12.1.1 IPAddress

string

#### 12.1.2 PaginatedRequest

Instructs the server to return only a subset of the data found on the server

@param pageSize number of entries returned per request

@param pageNumber what page of the set to returned, determined by the slice [pageSize\*pageNumber

struct

|            |     |
|------------|-----|
| pageSize   | int |
| pageNumber | int |

#### 12.1.3 PaginationInfo

struct

|              |     |
|--------------|-----|
| totalEntries | int |
|--------------|-----|

the total number of entries on the server, before matching

#### 12.1.4 SocketAddress

Tuple with IP address and port

struct

|         |           |
|---------|-----------|
| address | IPAddress |
| port    | int       |

IP address, either IPv4 or IPv6

#### 12.1.5 UUID

string

## 13 gatewayValidate (1.0)

### 13.1 Type Reference

#### 13.1.1 BigIntRange

##### struct

|     |               |
|-----|---------------|
| min | <b>bigint</b> |
| max | <b>bigint</b> |

#### 13.1.2 FloatRange

Float Range

##### struct

|     |                                            |
|-----|--------------------------------------------|
| min | <b>float</b><br>Minimum value of the range |
| max | <b>float</b><br>Maximum value of the range |

#### 13.1.3 IntRange

Integer Range

##### struct

|     |                                          |
|-----|------------------------------------------|
| min | <b>int</b><br>Minimum value of the range |
| max | <b>int</b><br>Maximum value of the range |

#### 13.1.4 Result

Validation result

##### struct

|        |                                                                                          |
|--------|------------------------------------------------------------------------------------------|
| result | <b>bool</b><br>True if config is valid and false otherwise.                              |
| msg    | <b>optional string</b><br>An error message, which may be given if the result is invalid. |

## 14 genericDescriptor (1.2)

### 14.1 Overview

Changelog:

#### 1.2

- **Changed**
  - The imported component version was bumped to 1.5.

#### 1.1

- **Changed**
  - The imported component version was bumped to 1.4.

### 14.2 Type Reference

#### 14.2.1 DescriptorAction

Dictate what action is to be performed on the provided descriptor.

**enum**

ADD  
REMOVE  
CHANGE

#### 14.2.2 GenericPmtDescriptorEntry

Defines a generic descriptor entry with a given action, tag and optional data.

|                  |                                                                  |
|------------------|------------------------------------------------------------------|
| @param action    | What action is to be performed with this descriptor entry.       |
| @param tag       | The Descriptor tag.                                              |
| @param component | Optional component (inner loop) this descriptor should apply to. |
| @param data      | Optional data to add to/replace the descriptor.                  |

**struct**

|           |                                             |
|-----------|---------------------------------------------|
| action    | <b>DescriptorAction</b>                     |
| tag       | <b>int</b>                                  |
| component | <b>optional component.ComponentSelector</b> |
| payload   | <b>optional string</b>                      |

## 15 outputMapping (1.5)

### 15.1 Overview

#### Changelog:

##### 1.5

- **Added**
  - Added field `sptsMode` to `OutputMappingTsWhitelist`
  - Added `SptsSource`, `SptsSettings` structs

##### 1.4

- **Changed**
  - componentMapping bumped from 1.2 to 1.3
  - genericDescriptor bumped from 1.1 to 1.2

##### 1.3

- **Added**
  - Added autoEmm option to TsSettings

##### 1.2

- **Added**
  - Added floating switch mode to InputRedundancy

##### 1.1

- **Changed**
  - genericDescriptor bumped to v 1.1

##### 1.0

- **Added**
  - Added mapping part from output.1.24.idcl
  - The field `externalPids` was added to `OutputMappingTsBlacklist`

### 15.2 Type Reference

#### 15.2.1 AnnexA

QAM Annex A input parameters, per the ITU-T J.83 specification.

By setting the Annex A parameters, one specifies how a TS should be QAMed, which ultimately results in a specific bitrate as described in the ITU-T spec.

#### struct

modulation

#### Modulation

Specifies the QAM modulation scheme, either QAM64 or QAM256.

symbolrate

#### Symbolrate

Symbol rate to use for QAM.

### 15.2.2 AnnexB

Annex B input parameters, per the ITU-T J.83 specification.

In contrast to Annex A, Annex B only has two choices for how to QAM, and therefore only has two possible output bitrates.

#### struct

modulation

#### Modulation

Specifies QAM modulation scheme, either QAM64 or QAM256. QAM64 results in a output bitrate of 26.970377 Mbps. QAM256 results in a output bitrate of 38.810700 Mbps.

### 15.2.3 AnnexC

QAM Annex C input parameters, per the ITU-T J.83 specification.

By setting the Annex C parameters, one specifies how a TS should be QAMed, which ultimately results in a specific bitrate as described in the ITU-T spec. Note that the resulting bitrate is the same as for Annex A.

#### struct

modulation

#### Modulation

Specifies the QAM modulation scheme, either QAM64 or QAM256.

symbolrate

#### Symbolrate

Symbol rate to use for QAM.

### 15.2.4 AtscChannelNumber

Major and minor channel overrides

#### struct

majorChannelNumber

#### int

Override major channel number.

minorChannelNumber

#### int

Override minor channel number.

### 15.2.5 AtscChannelSource

Source selection for a specific input service

#### struct

source

#### UUID

UUID for a input service.

inputRedundancy

#### InputRedundancy

priority

#### int

Value to identify the impact of losing specific source for redundancy logic

settings

#### AtscMultiplexChannelSettings

PID, PMT, CVCT and TVCT filters and overrides.

### 15.2.6 AtscMultiplexChannelSettings

PID, PMT, and SDT filters and overrides.

#### struct

|                     |                                                                                                                                                                        |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| programNumber       | <b>optional int</b>                                                                                                                                                    |
| channelNumber       | <b>optional AtscChannelNumber</b><br>Override channel number.                                                                                                          |
| shortName           | <b>optional string</b><br>Override short name.                                                                                                                         |
| componentLayout     | <b>componentMapping.ComponentLayoutMap</b><br>Remap, copy, and filter PIDs and components.                                                                             |
| pmtPid              | <b>optional int</b><br>Override PMT PID.                                                                                                                               |
| componentGeneration | <b>list of componentMapping.ComponentGeneration</b>                                                                                                                    |
| descriptors         | <b>list of genericDescriptor.GenericPmtDescriptorEntry</b><br>Add, change or remove descriptors on this service.                                                       |
| scramblingIndex     | <b>optional int</b><br>Scrambling index. May be given if scrambling is enabled for the TS, and this services should have different settings than the default settings. |

### 15.2.7 AtscMultiplexSource

ATSC multiplex source type, which must be either be a channel, an EMM or an external PID. An empty AtscMultiplexSource is considered an invalid config.

#### variant

|             |                                                       |
|-------------|-------------------------------------------------------|
| channel     | <b>AtscChannelSource</b><br>An ATSC channel (service) |
| emm         | <b>EmmSource</b><br>An EMM source                     |
| externalPid | <b>ExternalPidMapping</b><br>An External PID          |

### 15.2.8 AtscSource

#### variant

|           |                                    |
|-----------|------------------------------------|
| multiplex | <b>list of AtscMultiplexSource</b> |
|-----------|------------------------------------|

### 15.2.9 AtscTableSettings

ATSC PSI Table Settings

Settings to choose playout of PSI tables

#### struct

|      |                                                     |
|------|-----------------------------------------------------|
| cvct | <b>bool</b><br>Set playout of CVCT enabled/disabled |
|------|-----------------------------------------------------|

|      |                                                    |
|------|----------------------------------------------------|
| tvct | <b>bool</b><br>Set payout of TVCT enabled/disabled |
|------|----------------------------------------------------|

### 15.2.10 Bitrate

Bitrate describes the bitrate scheme to be used. Only one of the parameters can be specified.

#### variant

|        |                                                                                                           |
|--------|-----------------------------------------------------------------------------------------------------------|
| manual | <b>int</b><br>An integer that sets the desired bitrate in kbps (max 3000Mbps).                            |
| annexA | <b>AnnexA</b><br>A special struct for ITU-T J.83 Annex A, which contains modulation type and symbol rate. |
| annexB | <b>AnnexB</b><br>A special struct for ITU-T J.83 Annex B, which contains modulation type.                 |
| annexC | <b>AnnexC</b><br>A special struct for ITU-T J.83 Annex C, which contains modulation and symbol rate.      |
| vsb8   | <b>Vsb8</b><br>Used to indicate that the specific VSB-8 bitrate is to be used.                            |

### 15.2.11 DvbMultiplexServiceSettings

PID, PMT, and SDT filters and overrides.

#### struct

|                     |                                                                                                                                                                       |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| serviceId           | <b>optional int</b><br>Override service ID.                                                                                                                           |
| serviceName         | <b>optional string</b><br>Override service name.                                                                                                                      |
| serviceType         | <b>optional int</b><br>Override service type.                                                                                                                         |
| serviceProvider     | <b>optional string</b><br>Override service provider.                                                                                                                  |
| componentLayout     | <b>componentMapping.ComponentLayoutMap</b><br>Remap, copy, and filter PIDs and components.                                                                            |
| pmtPid              | <b>optional int</b><br>Override PMT PID.                                                                                                                              |
| componentGeneration | <b>list of componentMapping.ComponentGeneration</b>                                                                                                                   |
| descriptors         | <b>list of genericDescriptor.GenericPmtDescriptorEntry</b><br>Add, change or remove descriptors on this service.                                                      |
| scramblingIndex     | <b>optional int</b><br>Scrambling index. May be given if scrambling is enabled for the TS, and this service should have different settings than the default settings. |

### 15.2.12 DvbMultiplexSource

DVB multiplex source type, which must be either be a service, an EMM or an external PID. An empty DvbMultiplexSource is considered an invalid config.

#### variant

|             |                                              |
|-------------|----------------------------------------------|
| service     | <b>DvbServiceSource</b><br>An DVB service    |
| emm         | <b>EmmSource</b><br>An EMM source            |
| externalPid | <b>ExternalPidMapping</b><br>An External PID |

### 15.2.13 DvbServiceSource

Source selection for a specific input service

#### struct

|                 |                                                                                                                                                                              |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| source          | <b>UUID</b><br>UUID for a input service. The service ID can be queried using services:1.0/GetInputServices or calculated: UUID5( Input UUID, "dvbsource service SERVICEID"). |
| inputRedundancy | <b>InputRedundancy</b><br>is Input Redundancy configuration for this output                                                                                                  |
| priority        | <b>int</b><br>Value to identify the impact of losing specific source for redundancy logic                                                                                    |
| settings        | <b>DvbMultiplexServiceSettings</b><br>PID, PMT, and SDT filters and overrides.                                                                                               |

### 15.2.14 DvbSource

#### variant

|           |                                          |
|-----------|------------------------------------------|
| multiplex | <b>list</b> of <b>DvbMultiplexSource</b> |
|-----------|------------------------------------------|

### 15.2.15 EmmSource

EMM source

#### struct

|                 |                                                                                           |
|-----------------|-------------------------------------------------------------------------------------------|
| source          | <b>UUID</b><br>UUID of an EMM                                                             |
| inputRedundancy | <b>InputRedundancy</b><br>is Input Redundancy configuration for this output               |
| priority        | <b>int</b><br>Value to identify the impact of losing specific source for redundancy logic |

### 15.2.16 ExternalPidMapping

Map specific pid from a source with the specified uuid to specific destination pid on output

#### struct

|          |                                                                                           |
|----------|-------------------------------------------------------------------------------------------|
| source   | <b>UUID</b><br>UUID to identify which input source to map pid from                        |
| priority | <b>int</b><br>Value to identify the impact of losing specific source for redundancy logic |
| srcPid   | <b>int</b><br>PID to map from the specified source                                        |
| dstPid   | <b>int</b><br>PID to map the source PID to on the output                                  |

### 15.2.17 HotStandby

#### struct

|                  |                                                                                                                                                                                      |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| identicalSources | <b>optional IdenticalSources</b><br>When set MAIN and first BACKUP sources will be switched internally in FPGA module. MAIN and first BACKUP sources should have identical contents. |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 15.2.18 IdenticalSources

#### struct

|               |                                                                                                                                                                                                  |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| bufferDelay   | <b>int</b><br>Delay in milliseconds for offset handling between streams.                                                                                                                         |
| guardInterval | <b>int</b><br>Delay in milliseconds used to compensate for fluctuations in the time offset between the flows                                                                                     |
| switchOnTei   | <b>bool</b><br>When enabled FPGA module additionally uses TEI status to assess alternative health. Available only for HotStandbyWith-IdenticalSources switch mode. Only one alternative allowed. |
| latencyTuning | <b>LatencyTuning</b><br>Selected latency tuning strategy                                                                                                                                         |

### 15.2.19 InputRedundancy

Data structure related to Input Redundancy feature

#### struct

|         |                                                                                       |
|---------|---------------------------------------------------------------------------------------|
| mode    | <b>InputRedundancySwitchMode</b><br>Selected switch mode                              |
| backups | <b>list of UUID</b><br>List of alternative sources<br><b>optional PreferredSource</b> |

|                 |                                                                                                                                                                        |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| preferredSource | When set allows to override default source preferences                                                                                                                 |
| switchbackDelay | <b>int</b><br>Delay in seconds to wait before switching back to a more preferred alternative or MAIN source.                                                           |
| hotStandby      | <b>optional HotStandby</b><br>When set all alternative's bitrates will be monitored. Flows are preemptively mapped to the output card providing improved switch times. |

### 15.2.20 InputRedundancySwitchMode

Enum related to SwitchMode between alternative sources

#### enum

|           |                                                                                                                                                                                        |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Disabled  | InputRedundancy is inactive; only MAIN source is used.                                                                                                                                 |
| Reverting | Always switch to the highest priority, healthy alternative. Usually after it has been healthy for a set SWITCHBACK DELAY. Main has the highest priority, then backup 1, backup 2, etc. |
| Floating  | As opposed to reverting, it doesn't go back to a higher priority alternative even if it becomes healthy again after a SWITCHBACK DELAY.                                                |

### 15.2.21 LatencyTuning

Enum describing desired latency tuning strategy

#### enum

|             |                                                        |
|-------------|--------------------------------------------------------|
| Auto        | The first flow activated will be assumed to be leading |
| MainLeading | The Main source is assumed to be leading               |
| MainLagging | The Main source is assumed to be lagging               |

### 15.2.22 Modulation

QAM Modulation Types.

The J.83 specification utilizes two QAM variants, QAM64 and QAM256. It is denoted as "64-QAM" and "256-QAM" in the ITU-T J.83 spec. The number denotes the amount of points in the signal constellation used when performing QAM.

#### enum

|        |
|--------|
| QAM64  |
| QAM256 |

### 15.2.23 MpegMultiplexProgramSettings

PID and PMT filters and overrides.

#### struct

|               |                                                                                  |
|---------------|----------------------------------------------------------------------------------|
| programNumber | <b>optional int</b><br>Override program number (Equivalent to serviceId in DVB). |
|---------------|----------------------------------------------------------------------------------|

|                     |                                                                                                                                                                        |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| componentLayout     | <b>componentMapping.ComponentLayoutMap</b><br>Remap, copy and filter PIDs and components.                                                                              |
| pmtPid              | <b>optional int</b><br>Override the PMT PID.                                                                                                                           |
| componentGeneration | <b>list of componentMapping.ComponentGeneration</b>                                                                                                                    |
| descriptors         | <b>list of genericDescriptor.GenericPmtDescriptorEntry</b><br>Add, change or remove descriptors on this service.                                                       |
| scramblingIndex     | <b>optional int</b><br>Scrambling index. May be given if scrambling is enabled for the TS, and this services should have different settings than the default settings. |

### 15.2.24 MpegMultiplexSource

MPEG multiplex source type, which must be either be a program, an EMM or an external PID. An empty MpegMultiplexSource is considered an invalid config.

#### variant

|             |                                                       |
|-------------|-------------------------------------------------------|
| program     | <b>MpegProgramSource</b><br>An MPEG program (service) |
| emm         | <b>EmmSource</b><br>An EMM source                     |
| externalPid | <b>ExternalPidMapping</b><br>An External PID          |

### 15.2.25 MpegProgramSource

MPEG program source and settings bundle.

#### struct

|                 |                                                                                            |
|-----------------|--------------------------------------------------------------------------------------------|
| source          | <b>UUID</b><br>UUID of an input program (MPEG equivalent of DVB service)                   |
| inputRedundancy | <b>InputRedundancy</b><br>is Input Redundancy configuration for this output                |
| priority        | <b>int</b><br>Value to identify the impact of losing specific source for redundancy logic  |
| settings        | <b>MpegMultiplexProgramSettings</b><br>PID and PMT filters and overrides for this program. |

### 15.2.26 MpegSource

An MPEG mode source. Can be SPTS or MPTS, where SPTS is considered a single program multiplex struct.

#### variant

|           |                                                                                       |
|-----------|---------------------------------------------------------------------------------------|
| multiplex | <b>list of MpegMultiplexSource</b><br>A list of MPEG sources this output consists of. |
|-----------|---------------------------------------------------------------------------------------|

### 15.2.27 OutputMapping

Output settings variant, contains settings for the supported output types. It must be either tsWhitelistMode, tsBlacklistMode or rawMode. An empty OutputMapping is considered an invalid config.

#### variant

|                 |                                                                                           |
|-----------------|-------------------------------------------------------------------------------------------|
| tsWhitelistMode | <b>OutputMappingTsWhitelist</b><br>Build TS by adding content                             |
| tsBlacklistMode | <b>OutputMappingTsBlacklist</b><br>Map entire TS, and possibly exclude part of the stream |
| rawMode         | <b>OutputMappingRaw</b><br>Transparently map entire flow                                  |

### 15.2.28 OutputMappingAtsc

ATSC MPTS/SPTS Settings.

The system will generate PAT, PMT, MGT, and possibly CVCT and TVCT

#### struct

|          |                                                |
|----------|------------------------------------------------|
| settings | <b>TsSettings</b><br>Generic stream settings.  |
| source   | <b>AtscSource</b><br>Content to multiplex.     |
| tables   | <b>AtscTableSettings</b><br>PSI table settings |

### 15.2.29 OutputMappingDvb

DVB MPTS/SPTS Settings.

The system will generate PAT, PMT, and SDT

#### struct

|              |                                                             |
|--------------|-------------------------------------------------------------|
| orgNetworkId | <b>optional int</b><br>Orginal Network Id to set on stream. |
| settings     | <b>TsSettings</b><br>Generic stream settings.               |
| source       | <b>DvbSource</b><br>Content to multiplex.                   |

### 15.2.30 OutputMappingMpeg

#### struct

|          |                   |
|----------|-------------------|
| settings | <b>TsSettings</b> |
| source   | <b>MpegSource</b> |

### 15.2.31 OutputMappingRaw

Mapping of stream as is

#### struct

|                 |                                                                          |
|-----------------|--------------------------------------------------------------------------|
| source          | <b>UUID</b><br>Main source of stream                                     |
| inputRedundancy | <b>InputRedundancy</b><br>Input Redundancy configuration for this output |

### 15.2.32 OutputMappingSpts

SPTS Output Settings.

The system will generate PAT, PMT, and SDT

#### struct

|              |                                                              |
|--------------|--------------------------------------------------------------|
| orgNetworkId | <b>optional int</b><br>Original Network Id to set on stream. |
| settings     | <b>TsSettings</b><br>Generic stream settings.                |
| source       | <b>SptsSource</b><br>Content to output                       |

### 15.2.33 OutputMappingTsBlacklist

Transport stream blacklist mapping

If CBR is specified, the original NULL PIDs are discarded.

#### struct

|                 |                                                                                                                                          |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------|
| source          | <b>UUID</b><br>Main TS source                                                                                                            |
| inputRedundancy | <b>InputRedundancy</b><br>Input Redundancy configuration for this output                                                                 |
| bitrate         | <b>Bitrate</b><br>Variant Bitrate - if specified, the transport is CBR, AnnexA or AnnexB with relevant attributes. Otherwise, it is VBR. |
| stopPids        | <b>list of int</b><br>PIDs to be stopped on Playout                                                                                      |
| remapPids       | <b>map from int to int</b><br>PIDs to be remapped on Playout                                                                             |
| externalPids    | <b>list of ExternalPidMapping</b><br>List of PID mappings from external sources                                                          |

### 15.2.34 OutputMappingTsWhitelist

Transport stream whitelist mapping which must be either DVB, MPEG, ATSC or SPTS (only supported on 100G product). An empty OutputMappingTsWhitelist is considered an invalid config.

#### variant

|          |                                                                                                                 |
|----------|-----------------------------------------------------------------------------------------------------------------|
| mpegMode | <b>OutputMappingMpeg</b><br>MPEG mode will create an output with MPEG MPTS/SPTS settings                        |
| dvbMode  | <b>OutputMappingDvb</b><br>DVB mode will create an output with DVB MPTS/SPTS settings                           |
| atscMode | <b>OutputMappingAtsc</b><br>ATSC mode will create an output with ATSC MPTS/SPTS settings                        |
| sptsMode | <b>OutputMappingSpts</b><br>SPTS mode will create an SPTS output. Currently only supported on the 100G product. |

### 15.2.35 PreferredSource

Preferred source configuration

#### struct

|        |                                                                                                                                                                                                                                                                                                                                                       |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| index  | <b>int</b><br>Selects a source that should be marked as preferred. Index = 0 is MAIN source, index = 1 means first backup, etc. Preferred source will be used as the default source when it is healthy. If it fails the system will switch using regular source priorities (for example: first MAIN, then first backup, then second backup and so on) |
| locked | <b>bool</b><br>When enabled disables switching to another source when preferred source is faulty                                                                                                                                                                                                                                                      |

### 15.2.36 SptsSettings

PID, PMT, and SDT filters and overrides.

#### struct

|                 |                                                      |
|-----------------|------------------------------------------------------|
| serviceId       | <b>optional int</b><br>Override service ID.          |
| serviceName     | <b>optional string</b><br>Override service name.     |
| serviceType     | <b>optional int</b><br>Override service type.        |
| serviceProvider | <b>optional string</b><br>Override service provider. |
| pmtPid          | <b>optional int</b><br>Override PMT PID.             |

### 15.2.37 SptsSource

SPTS source and settings bundle.

#### struct

**UUID**

|                 |                                                                             |
|-----------------|-----------------------------------------------------------------------------|
| source          | UUID of an input SPTS source                                                |
| inputRedundancy | <b>InputRedundancy</b><br>is Input Redundancy configuration for this output |
| sptsSettings    | <b>SptsSettings</b><br>PID, PMT, and SDT filters and overrides.             |

### 15.2.38 Symbolrate

Symbol rate.

A symbol rate is defined in DOCSIS 2.0 as  $(m/n) * 10.24\text{MHz}$ , where M and N are 16-bit unsigned, nonzero integers, and 10.24MHz is the CMTS clock frequency.

#### struct

|   |                                                                                |
|---|--------------------------------------------------------------------------------|
| m | <b>int</b><br>The numerator of the desired fraction of CMTS clock frequency.   |
| n | <b>int</b><br>The denominator of the desired fraction of CMTS clock frequency. |

### 15.2.39 TsSettings

Settings shared between all TS configurations

#### struct

|         |                                                                                                                                                                                                                                                                                            |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| tsId    | <b>int</b><br>Transport stream ID for the output TS                                                                                                                                                                                                                                        |
| bitrate | <b>Bitrate</b><br>Variant Bitrate - if specified, the transport is CBR, AnnexA or AnnexB with relevant attributes. Otherwise, it is VBR.                                                                                                                                                   |
| autoEmm | <b>bool</b><br>Automatically include CAT and EMM from the input source. Automatic CAT/EMM is supported for single service outputs (SPTS) only. Also, only the CAS ID referenced in the ECM for the service will be mapped. The CAT will be regenerated to signal the required CAS ID only. |

### 15.2.40 Vsb8

8-VSB struct. It is used by the Bitrate type to specify that the 8-VSB modulation scheme is desired. Per the specification, 8-VSB has no configurable inputs, and returns a MPEG II stream of 19.39...Mbps/s ready for 8-VSB modulation.

empty **struct**

## 16 s2xTypes (1.0)

### 16.1 Type Reference

#### 16.1.1 CodeRate

Code Rate

enum

|          |                                                                                         |
|----------|-----------------------------------------------------------------------------------------|
| F11_45   | Fraction 11/45                                                                          |
| F1_4     | Fraction 1/4                                                                            |
| F4_15    | Fraction 4/15                                                                           |
| F13_45   | Fraction 13/45                                                                          |
| F14_45   | Fraction 14/45                                                                          |
| F1_3     | Fraction 1/3                                                                            |
| F2_5     | Fraction 2/5                                                                            |
| F9_20    | Fraction 9/20                                                                           |
| F7_15    | Fraction 7/15                                                                           |
| F1_2     | Fraction 1/2                                                                            |
| F1_2_L   | Fraction 1/2                                                                            |
| F8_15    | Fraction 8/15                                                                           |
| F8_15_L  | Fraction 8/15                                                                           |
| F11_20   | Fraction 11/20                                                                          |
| F5_9_L   | Fraction 5/9                                                                            |
| F26_45   | Fraction 26/45                                                                          |
| F26_45_L | Fraction 26/45                                                                          |
| F3_5     | Fraction 3/5                                                                            |
| F3_5_L   | Fraction 3/5                                                                            |
| F28_45   | Fraction 28/45                                                                          |
| F23_36   | Fraction 23/36                                                                          |
| F29_45_L | Fraction 29/45                                                                          |
| F2_3     | Fraction 2/3                                                                            |
| F2_3_L   | Fraction 2/3                                                                            |
| F31_45_L | Fraction 31/45                                                                          |
| F25_36   | Fraction 25/36                                                                          |
| F32_45   | Fraction 32/45                                                                          |
| F32_45_L | Fraction 32/45                                                                          |
| F13_18   | Fraction 13/18                                                                          |
| F11_15   | Fraction 11/15                                                                          |
| F11_15_L | Fraction 11/15                                                                          |
| F3_4     | Fraction 3/4                                                                            |
| F7_9     | Fraction 7/9                                                                            |
| F4_5     | Fraction 4/5                                                                            |
| F5_6     | Fraction 5/6                                                                            |
| F77_90   | Fraction 77/90                                                                          |
| F6_7     | Fraction 6/7; only detected Code Rate in Demodulator status,<br>not valid for Modulator |
| F7_8     | Fraction 7/8                                                                            |
| F8_9     | Fraction 8/9                                                                            |

|       |               |
|-------|---------------|
| F9_10 | Fraction 9/10 |
|-------|---------------|

### 16.1.2 Constellation

Constellation

#### enum

|         |         |
|---------|---------|
| QPSK    | QPSK    |
| PSK8    | PSK8    |
| APSK8   | APSK8   |
| APSK16  | APSK16  |
| APSK32  | APSK32  |
| APSK64  | APSK64  |
| APSK128 | APSK128 |
| APSK256 | APSK256 |

### 16.1.3 FrameLength

FEC Frame Length

#### enum

|        |                     |
|--------|---------------------|
| SHORT  | Short frame length  |
| NORMAL | Normal frame length |

### 16.1.4 PIMode

Physical Layer Scrambling Mode

#### enum

|               |                                                               |
|---------------|---------------------------------------------------------------|
| PL_SEQUENCE   | Preferred scrambling sequence as defined in ETSI EN 302 307-2 |
| GOLD_SEQUENCE | Gold Sequence Index                                           |
| PL_ROOT       | 18-bit Pseudorandom Binary Sequence (PRBS) root               |

### 16.1.5 RollOff

Roll-off

All values are allowed in S1, S2 and S2X mode for simplicity.

#### enum

|       |                     |
|-------|---------------------|
| D0_05 | Decimal number 0.05 |
| D0_10 | Decimal number 0.10 |
| D0_15 | Decimal number 0.15 |
| D0_20 | Decimal number 0.20 |
| D0_25 | Decimal number 0.25 |
| D0_35 | Decimal number 0.35 |

## 16.1.6 System

System (DVB Standard)

### enum

|     |                  |
|-----|------------------|
| S1  | DVB-S standard   |
| S2  | DVB-S2 standard  |
| S2X | DVB-S2X standard |

## 17 s2xValidate (1.0)

### 17.1 Type Reference

#### 17.1.1 FloatRange

Float Range

##### struct

|     |                                            |
|-----|--------------------------------------------|
| min | <b>float</b><br>Minimum value of the range |
| max | <b>float</b><br>Maximum value of the range |

#### 17.1.2 IntRange

Integer Range

##### struct

|     |                                          |
|-----|------------------------------------------|
| min | <b>int</b><br>Minimum value of the range |
| max | <b>int</b><br>Maximum value of the range |

#### 17.1.3 Result

Validation result

##### struct

|        |                                                                                          |
|--------|------------------------------------------------------------------------------------------|
| result | <b>bool</b><br>True if config is valid and false otherwise.                              |
| msg    | <b>optional string</b><br>An error message, which may be given if the result is invalid. |

## 18 s2xoutBissSettings (1.0)

## 19 s2xoutCarrierId (1.0)

### 19.1 Command Reference

#### 19.1.1 GetCarrierIds

Get CarrierId objects from database.

- message **GetCarrierIds.Request**
- message **GetCarrierIds.Response**

#### 19.1.2 SetCarrierIds

Create/update CarrierId objects in database.

- message **SetCarrierIds.Request**
- message **SetCarrierIds.Response**

#### 19.1.3 DeleteCarrierIds

Delete CarrierId objects from database.

- message **DeleteCarrierIds.Request**
- message **DeleteCarrierIds.Response**

### 19.2 Type Reference

#### 19.2.1 CarrierId

Carrier Identification (CID) configuration

##### struct

|           |                             |
|-----------|-----------------------------|
| label     | <b>string</b><br>User label |
| latitude  | <b>string</b><br>Latitude   |
| longitude | <b>string</b><br>Longitude  |
| telephone | <b>string</b><br>Telephone  |
| userData  | <b>string</b><br>User data  |

#### 19.2.2 DeleteCarrierIds.Request

##### struct

|     |                     |
|-----|---------------------|
| ids | <b>list of UUID</b> |
|-----|---------------------|

### 19.2.3 DeleteCarrierIds.Response

empty **struct**

### 19.2.4 GetCarrierIds.Request

empty **struct**

### 19.2.5 GetCarrierIds.Response

**struct**

data **map** from **UUID** to **CarrierId**

### 19.2.6 SetCarrierIds.Request

**struct**

data **map** from **UUID** to **CarrierId**

### 19.2.7 SetCarrierIds.Response

empty **struct**

## 20 s2xoutModulator (1.13)

### 20.1 Overview

#### Changelog

##### 1.13

##### Changed

- outputMapping from 1.4 to 1.5

##### 1.12

##### Changed

- bissScrambling from 1.5 to 1.6

##### 1.11

##### Changed

- outputMapping from 1.3 to 1.4

##### 1.10

- no changes in API
- non functional changes to internal implementation

### 20.2 Command Reference

#### 20.2.1 GetModulators

Get Modulator objects from database.

- message **GetModulators.Request**
- message **GetModulators.Response**

#### 20.2.2 SetModulators

Update Modulator objects in database.

Note: This interface should **NOT** be used to create any new Modulator objects. Instead, only use it to modify existing objects.

- message **SetModulators.Request**
- message **SetModulators.Response**

#### 20.2.3 DeleteModulators

Delete Modulator objects from database.

Note: This command should **NOT** be used (except for debugging purposes). Modulator objects should never be deleted through the API.

- message **DeleteModulators.Request**
- message **DeleteModulators.Response**

### 20.3 Type Reference

### 20.3.1 ComponentGroups

Component Groups

High level abstraction groups for the elementary streams.

#### enum

|             |                                        |
|-------------|----------------------------------------|
| ALL         | Audio / video / teletext and subtitles |
| AUDIO_VIDEO | Audio / video                          |
| VIDEO       | Video                                  |
| NONE        | None                                   |

### 20.3.2 DeleteModulators.Request

#### struct

ids **list** of **UUID**

### 20.3.3 DeleteModulators.Response

empty **struct**

### 20.3.4 GetModulators.Request

empty **struct**

### 20.3.5 GetModulators.Response

#### struct

data **map** from **UUID** to **Modulator**

### 20.3.6 ModulationOverrides

Override Modulation Settings configured in Port Profile

The modulation settings are configured in Port Profile, some of these values may be overridden.

#### struct

|               |                                                                      |
|---------------|----------------------------------------------------------------------|
| system        | <b>optional s2xTypes.System</b><br>DVB S1 / S2 / S2X system          |
| frameLength   | <b>optional s2xTypes.FrameLength</b><br>Length of FEC frame          |
| constellation | <b>optional s2xTypes.Constellation</b><br>Constellation diagram type |
| codeRate      | <b>optional s2xTypes.CodeRate</b><br>Code rate                       |

### 20.3.7 Modulator

#### Modulator Settings

Configurations related to the modulation of a single TS stream.

##### struct

|                     |                                                                            |
|---------------------|----------------------------------------------------------------------------|
| port                | <b>s2xoutPhysicalPort.Port</b><br>Physical port this output belongs to     |
| label               | <b>string</b><br>User defined label                                        |
| overrides           | <b>ModulationOverrides</b><br>Override modulation settings of profile      |
| source              | <b>outputMapping.OutputMapping</b><br>Content sent out on this modulator   |
| scrambling          | <b>bissScrambling.ScramblerSettings</b><br>Scrambling of TS data           |
| scrambledComponents | <b>ComponentGroups</b><br>Scrambled component groups for scrambled streams |

### 20.3.8 SetModulators.Request

##### struct

data **map** from **UUID** to **Modulator**

### 20.3.9 SetModulators.Response

empty **struct**

## 21 s2xoutPhysicalPort (1.3)

### 21.1 Overview

#### Changelog

#### 1.3

#### Added

- a new field `bucFrequency` in `PhysicalPort` type (see **Satellite mode** section in **Overview** of `s2xout`)

### 21.2 Command Reference

#### 21.2.1 GetPhysicalPorts

Get `PhysicalPort` objects from database.

- message `GetPhysicalPorts.Request`
- message `GetPhysicalPorts.Response`

#### 21.2.2 SetPhysicalPorts

Update `PhysicalPort` objects in database.

Note: This interface should **NOT** be used to create any new `PhysicalPort` objects. Instead, only use it to modify existing objects.

- message `SetPhysicalPorts.Request`
- message `SetPhysicalPorts.Response`

#### 21.2.3 ResetPortSwitchback

Resets redundancy switched status on `PhysicalPort` Allows ports with switchback disabled to switch back to MAIN.

- message `ResetPortSwitchback.Request`
- message `ResetPortSwitchback.Response`

#### 21.2.4 DeletePhysicalPorts

Delete `PhysicalPort` objects from database.

Note: This command should **NOT** be used (except for debugging purposes). `PhysicalPort` objects should never be deleted through the API.

- message `DeletePhysicalPorts.Request`
- message `DeletePhysicalPorts.Response`

### 21.3 Type Reference

#### 21.3.1 DeletePhysicalPorts.Request

**struct**

`ids`

list of **UUID**

### 21.3.2 DeletePhysicalPorts.Response

empty **struct**

### 21.3.3 EffectiveOutputConfig

**struct**

|                  |                                      |
|------------------|--------------------------------------|
| label            | string                               |
| rf               | EffectiveRfSettings                  |
| modulation       | s2xoutPortProfile.ModulationSettings |
| linearGain       | optional UUID                        |
| linearGroupDelay | optional UUID                        |
| carrierId        | optional UUID                        |

### 21.3.4 EffectiveRfSettings

**struct**

|                  |                              |
|------------------|------------------------------|
| outputMode       | s2xoutPortProfile.OutputMode |
| frequency        | float                        |
| rfLevel          | float                        |
| symbolRate       | float                        |
| rolloff          | s2xTypes.Rolloff             |
| invertedSpectrum | bool                         |

### 21.3.5 GetPhysicalPorts.Request

empty **struct**

### 21.3.6 GetPhysicalPorts.Response

**struct**

|      |                               |
|------|-------------------------------|
| data | map from UUID to PhysicalPort |
|------|-------------------------------|

### 21.3.7 MuteOnError

MuteOnError

**struct**

|                     |                                                                         |
|---------------------|-------------------------------------------------------------------------|
| switchbackThreshold | int<br>Threshold where the redundancy system unmutes the output         |
| autoUnmute          | bool<br>If true, the redundancy system automatically unmutes the output |

### 21.3.8 PhysicalPort

PhysicalPort configuration

#### struct

|                |                                                                                                                                                                                                                                 |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| port           | <b>Port</b><br>Physical port number                                                                                                                                                                                             |
| enabled        | <b>bool</b><br>Enable/disable output                                                                                                                                                                                            |
| redundancy     | <b>RedundancySwitch</b><br>Redundancy switch                                                                                                                                                                                    |
| redundancyMode | <b>RedundancyMode</b><br>Redundancy mode                                                                                                                                                                                        |
| profile        | <b>optional UUID</b><br>UUID reference to selected PortProfile                                                                                                                                                                  |
| overrides      | <b>RfOverrides</b><br>Override portProfile settings                                                                                                                                                                             |
| bucFrequency   | <b>optional float</b><br>If set then the system will automatically determine carrier frequency of the signal based on BUC and satellite transmission frequency. See <b>Satellite mode</b> section in <b>Overview</b> of s2xout. |
| refOutput10Mhz | <b>bool</b><br>Enable 10Mhz reference output                                                                                                                                                                                    |
| dcOutput24V    | <b>bool</b><br>Enable 24V DC output                                                                                                                                                                                             |
| cwCarrier      | <b>bool</b><br>Enable continuous wave mode (for debugging)                                                                                                                                                                      |

### 21.3.9 Port

Physical Output Port

The ports are numbered from left to right on the card.

#### enum

|    |        |
|----|--------|
| P1 | Port 1 |
| P2 | Port 2 |

### 21.3.10 RedundancyMode

Redundancy Mode action to perform on Error

#### variant

|        |                                                    |
|--------|----------------------------------------------------|
| mute   | <b>MuteOnError</b><br>Mute on Error                |
| backup | <b>SwitchToBackup</b><br>Switch to Backup on error |

### 21.3.11 RedundancySwitch

Redundancy Switch

Controls the redundancy switch from the software

#### enum

|        |                                                                                   |
|--------|-----------------------------------------------------------------------------------|
| AUTO   | Depending on RedundancyMode, the output will switch to backup or it will be muted |
| MAIN   | Force MAIN                                                                        |
| BACKUP | Force BACKUP                                                                      |
| MUTE   | Force MUTE                                                                        |

### 21.3.12 ResetPortSwitchback.Request

#### struct

ids [list of UUID](#)

### 21.3.13 ResetPortSwitchback.Response

empty [struct](#)

### 21.3.14 RfOverrides

Override RF Settings configured in Port Profile

The RF settings are configured in Port Profile, some of these values may be overridden.

#### struct

|            |                                                                                                                           |
|------------|---------------------------------------------------------------------------------------------------------------------------|
| frequency  | <a href="#">optional float</a><br>Carrier frequency or carrier satellite frequency (if BUC frequency is enabled) (in MHz) |
| rfLevel    | <a href="#">optional float</a><br>RF Level (in dBm)                                                                       |
| symbolRate | <a href="#">optional float</a><br>Symbol rate (in Mbd)                                                                    |

### 21.3.15 SetPhysicalPorts.Request

#### struct

data [map](#) from [UUID](#) to [PhysicalPort](#)

### 21.3.16 SetPhysicalPorts.Response

empty [struct](#)

### 21.3.17 SwitchToBackup

SwitchToBackup

#### struct

|                     |                                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| switchbackThreshold | <b>int</b><br>Threshold where the redundancy system switches back to Main         |
| backupThreshold     | <b>float</b><br>Threshold for a backup to register as present (in dBm)            |
| autoSwitchBack      | <b>bool</b><br>If true, the redundancy system automatically switches back to Main |

## 22 s2xoutPortProfile (1.1)

### 22.1 Overview

#### Changelog

##### 1.1

#### Added

- port profiles are now divided into two categories – direct and satellite
  - the distinction is based on RfSettings variant
  - see **Satellite mode** section in **Overview** of s2xout
  - see RfSettings, RfSettingsDirect, RfSettingsSatellite in **Type Reference** section of this module

#### Changed

- RfSettings does no longer directly contain RF settings but becomes a variant determining profile category
- Allow S2 constellations in S2X mode

### 22.2 Command Reference

#### 22.2.1 GetPortProfiles

Get PortProfile objects from database.

- message **GetPortProfiles.Request**
- message **GetPortProfiles.Response**

#### 22.2.2 SetPortProfiles

Create/update PortProfile objects in database.

- message **SetPortProfiles.Request**
- message **SetPortProfiles.Response**

#### 22.2.3 DeletePortProfiles

Delete PortProfile objects from database.

- message **DeletePortProfiles.Request**
- message **DeletePortProfiles.Response**

### 22.3 Type Reference

#### 22.3.1 DeletePortProfiles.Request

##### struct

ids [list of UUID](#)

### 22.3.2 DeletePortProfiles.Response

empty **struct**

### 22.3.3 GetPortProfiles.Request

empty **struct**

### 22.3.4 GetPortProfiles.Response

**struct**

data **map** from **UUID** to **PortProfile**

### 22.3.5 GoldCode

Gold Code

Gold Code Scrambling Sequence

**enum**

|     |             |
|-----|-------------|
| GC0 | Gold code 0 |
| GC1 | Gold code 1 |
| GC2 | Gold code 2 |
| GC3 | Gold code 3 |
| GC4 | Gold code 4 |
| GC5 | Gold code 5 |
| GC6 | Gold code 6 |

### 22.3.6 ModulationSettings

Modulation Settings

**struct**

|                  |                                                                    |
|------------------|--------------------------------------------------------------------|
| system           | <b>s2xTypes.System</b><br>DVB S1 / S2 / S2X system                 |
| frameLength      | <b>s2xTypes.FrameLength</b><br>Length of FEC frame                 |
| constellation    | <b>s2xTypes.Constellation</b><br>Constellation diagram type        |
| codeRate         | <b>s2xTypes.CodeRate</b><br>Code rate                              |
| pilot            | <b>bool</b><br>Enable pilot                                        |
| plScramblingMode | <b>s2xTypes.PIMode</b><br>Decides how PL scrambling is sent        |
| plScramblingCode | <b>int</b><br>PL scrambling, in unit according to plScramblingMode |

### 22.3.7 OutputMode

OutputMode

**enum**

IF  
L\_BAND

### 22.3.8 PortProfile

Output port profile

Profile settings for a transponder chain.

**struct**

|                  |                                                                 |
|------------------|-----------------------------------------------------------------|
| label            | <b>string</b><br>User label                                     |
| rf               | <b>RfSettings</b><br>RF settings                                |
| modulation       | <b>ModulationSettings</b><br>Modulation settings                |
| linearGain       | <b>optional UUID</b><br>Apply linear gain precorrection         |
| linearGroupDelay | <b>optional UUID</b><br>Apply linear group delay precorrection  |
| carrierId        | <b>optional UUID</b><br>Playout of carrier identification (CID) |

### 22.3.9 RfSettings

There are two ways in which RF settings can be configured and this variant defines which one is used.

**variant**

|           |                            |
|-----------|----------------------------|
| direct    | <b>RfSettingsDirect</b>    |
| satellite | <b>RfSettingsSatellite</b> |

### 22.3.10 RfSettingsDirect

This is the direct way of configuring RF settings. Carrier frequency configured here is the actual frequency used in the output signal from the card.

This way of configuring RF settings is allowed only for physical ports which have NOT enabled BUC frequency.

**struct**

|                 |                                            |
|-----------------|--------------------------------------------|
| outputMode      | <b>OutputMode</b><br>Output mode           |
| outputFrequency | <b>float</b><br>Carrier frequency (in MHz) |
|                 | <b>float</b>                               |

|                  |                                            |
|------------------|--------------------------------------------|
| rfLevel          | RF Level (in dBm)                          |
| symbolRate       | <b>float</b><br>Symbol rate (in Mbd)       |
| rolloff          | <b>s2xTypes.Rolloff</b><br>Roll-off factor |
| invertedSpectrum | <b>bool</b><br>Enable inverted spectrum    |

### 22.3.11 RfSettingsSatellite

This way of configuring RF settings is made to simplify process of configuring satellite uplink and allows to specify satellite transmission frequency.

See **Satellite mode** section in **Overview** of s2xout for more details.

This way of configuring RF settings is allowed only for physical ports which have enabled BUC frequency.

Useful details:

- satellite frequency – must be within L-band distance from BUC frequency configured on the physical port, i.e, BUC freq. +/- 950–2150Mhz
- output carrier frequency – automatically calculated based on configured satellite frequency and BUC frequency
- output mode – automatically set to L-band
- inverted spectrum – automatically determined (if satellite frequency is smaller than BUC frequency then spectrum will be inverted)

#### struct

|                    |                                                            |
|--------------------|------------------------------------------------------------|
| satelliteFrequency | <b>float</b><br>Satellite transmission frequency (in MHz). |
| rfLevel            | <b>float</b><br>RF Level (in dBm)                          |
| symbolRate         | <b>float</b><br>Symbol rate (in Mbd)                       |
| rolloff            | <b>s2xTypes.Rolloff</b><br>Roll-off factor                 |

### 22.3.12 SetPortProfiles.Request

#### struct

|      |                                                   |
|------|---------------------------------------------------|
| data | <b>map</b> from <b>UUID</b> to <b>PortProfile</b> |
|------|---------------------------------------------------|

### 22.3.13 SetPortProfiles.Response

empty **struct**

## 23 s2xoutPrecorrection (1.0)

### 23.1 Command Reference

#### 23.1.1 GetPrecorrections

Get Precorrection objects from database.

- message **GetPrecorrections.Request**
- message **GetPrecorrections.Response**

#### 23.1.2 SetPrecorrections

Create/update Precorrection objects in database.

- message **SetPrecorrections.Request**
- message **SetPrecorrections.Response**

#### 23.1.3 DeletePrecorrections

Delete Precorrection objects from database.

- message **DeletePrecorrections.Request**
- message **DeletePrecorrections.Response**

### 23.2 Type Reference

#### 23.2.1 DeletePrecorrections.Request

**struct**

ids **list** of **UUID**

#### 23.2.2 DeletePrecorrections.Response

empty **struct**

#### 23.2.3 GetPrecorrections.Request

empty **struct**

#### 23.2.4 GetPrecorrections.Response

**struct**

data **map** from **UUID** to **Precorrection**

#### 23.2.5 Graph

Two-dimensional graph

**struct**

points

list of **Point**

The points in the plane

### 23.2.6 Point

Point in the plane

**struct**

x

**float**

x-coordinate

y

**float**

y-coordinate

### 23.2.7 Precorrection

Precorrection

**struct**

label

**string**

User label

method

**PrecorrectionMethod**

Method of precorrection

graph

**Graph**

Precorrection graph specification

### 23.2.8 PrecorrectionMethod

Precorrection methods

**enum**

LINEAR\_GAIN

Linear gain precorrection x: Delta from carrier (MHz) y: Gain rel. to mean power (dB)

LINEAR\_GROUP\_DELAY

Linear group delay precorrection x: Delta from carrier (MHz) y: Delay (ns)

### 23.2.9 SetPrecorrections.Request

**struct**

data

map from **UUID** to **Precorrection**

### 23.2.10 SetPrecorrections.Response

empty **struct**

## 24 s2xoutStatus (1.3)

### 24.1 Overview

Changelog:

1.3

- Added
  - The type OutputSourceStatus was added.
  - The field outputSources was added to the RedundancyStatus type.

### 24.2 Command Reference

#### 24.2.1 GetPhysicalPortStatus

Get PhysicalPort status

- message **GetPhysicalPortStatus.Request**
- message **GetPhysicalPortStatus.Response**

### 24.3 Type Reference

#### 24.3.1 BitrateStatus

Bitrate Status

**struct**

|                |                                                                  |
|----------------|------------------------------------------------------------------|
| cbr            | <b>float</b><br>Current CBR of the saved configuration (in Mbps) |
| payload        | <b>float</b><br>Current payload (in Mbps)                        |
| maximumPayload | <b>float</b><br>Maximum Payload in the last 60 seconds (in Mbps) |

#### 24.3.2 GetPhysicalPortStatus.Request

**PhysicalPortStatusRequest**

#### 24.3.3 GetPhysicalPortStatus.Response

**PhysicalPortStatusResponse**

#### 24.3.4 HardwareSwitch

Position of the rotary switch on the front panel

**enum**

|        |              |
|--------|--------------|
| MAIN   | Force Main   |
| BACKUP | Force Backup |

|         |                                                        |
|---------|--------------------------------------------------------|
| AUTO    | Redundancy switch is controled by the RedundancySwitch |
| MUTE    | Force Mute                                             |
| FAILURE | Indicates a failure in the rotary switch               |

### 24.3.5 LedStatus

Status of the front panel LEDs

#### enum

|                       |                                                                                          |
|-----------------------|------------------------------------------------------------------------------------------|
| GREEN                 | Auto mode, Main selected; HardwareSwitch = AUTO, RedundancySwitch = MAIN                 |
| GREEN_BLINKING        | Force Main; HardwareSwitch = MAIN                                                        |
| BLUE                  | Auto mode, CPU dead or Backup selected; HardwareSwitch = AUTO, RedundancySwitch = BACKUP |
| BLUE_BLINKING         | Force Backup; HardwareSwitch = BACKUP                                                    |
| RED                   | No output; HardwareSwitch = AUTO, RedundancySwitch = MUTE                                |
| RED_BLINKING          | Force Mute; HardwareSwitch = MUTE                                                        |
| ROTARY_SWITCH_FAILURE | Rotary switch failure                                                                    |

### 24.3.6 MainStatus

Status of Main RF Source

#### enum

|      |
|------|
| GOOD |
| BAD  |

### 24.3.7 OutputSourceStatus

Output Source Status for Input Redundancy

#### struct

|                       |                                                                                                                          |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------|
| currentAlternativeIdx | <b>int</b><br>Index of currently selected alternative. 0 is MAIN, 1 is first backup and so on                            |
| alternativeStatuses   | <b>list</b> of <b>OutputSourceStatus.alternativeStatuses.item</b><br>Status of each alternative defined for given source |

### 24.3.8 OutputSourceStatus.alternativeStatuses.item

#### struct

|          |             |
|----------|-------------|
| isBroken | <b>bool</b> |
|----------|-------------|

### 24.3.9 PhysicalPortStatus

PhysicalPort status

**struct**

|            |                                                       |
|------------|-------------------------------------------------------|
| redundancy | <b>optional RedundancyStatus</b><br>Redundancy status |
| bitrates   | <b>optional BitrateStatus</b><br>Bitrate Status       |

**24.3.10 PhysicalPortStatusRequest**

PhysicalPort status request

**struct**

|     |                              |
|-----|------------------------------|
| ids | <b>optional list of UUID</b> |
|-----|------------------------------|

**24.3.11 PhysicalPortStatusResponse**

Response of a PhysicalPort status request

**struct**

|      |                                            |
|------|--------------------------------------------|
| data | <b>map from UUID to PhysicalPortStatus</b> |
|------|--------------------------------------------|

**24.3.12 RedundancyStatus**

Redundancy status

**struct**

|                |                                                                            |
|----------------|----------------------------------------------------------------------------|
| hardwareSwitch | <b>HardwareSwitch</b><br>Hardware switch position                          |
| rfSource       | <b>RfSource</b><br>Source in use                                           |
| led            | <b>LedStatus</b><br>Front plate LED status                                 |
| mainStatus     | <b>MainStatus</b><br>Status of Main RF Source                              |
| watchdogAlive  | <b>bool</b><br>Watchdog alive or dead                                      |
| backupPresent  | <b>optional bool</b><br>Backup input present                               |
| backupLevel    | <b>optional float</b><br>Level on the backup port (in dB)                  |
| outputSources  | <b>list of OutputSourceStatus</b><br>Status for each service in the output |

**24.3.13 RfSource**

Source in use

**enum**

|        |                                         |
|--------|-----------------------------------------|
| MAIN   | Output RF is generated on this module   |
| BACKUP | Input RF signal is routed from RF input |
| MUTE   | The RF output is muted                  |

## 25 streamtype (1.0)

### 25.1 Type Reference

#### 25.1.1 StreamType

StreamType as defined in ISO 13818-1 Table 2-29.

##### enum

|                         |  |
|-------------------------|--|
| MPEG_VIDEO              |  |
| MPEG_AUDIO              |  |
| TTX                     |  |
| DVBSUB                  |  |
| PRIVATE                 |  |
| AC_3                    |  |
| H264                    |  |
| VBI                     |  |
| AAC                     |  |
| VC_1                    |  |
| EAC3                    |  |
| AIT                     |  |
| HBBTV_CAROUSEL          |  |
| DATA_CAROUSEL           |  |
| H265                    |  |
| S302M                   |  |
| ATSC_MPEG2_VIDEO        |  |
| ATSC_AC3_AUDIO          |  |
| ATSC_SCTE_35            |  |
| ATSC_ENHANCED_AC3_AUDIO |  |

## 26 tsstatus (1.5)

### 26.1 Overview

Changelog:

#### 1.5

- Added
  - RPC `GetOutputPidStatus` was added, which takes in a list of uuid of outputs and returns the status of the respective output transport streams.

#### 1.4

- Changed
  - component from 1.4 to 1.5

#### 1.3

- Changed
  - Dependency module component bumped from version 1.1 to 1.4

### 26.2 Command Reference

#### 26.2.1 `GetOutputPidStatus`

Get output PID status

- message `GetOutputPidStatus.Request`
- message `GetOutputPidStatus.Response`

### 26.3 Type Reference

#### 26.3.1 `GetOutputPidStatus.Request`

`OutputPidStatusRequest`

#### 26.3.2 `GetOutputPidStatus.Response`

`OutputPidStatusResponse`

#### 26.3.3 `OutputPidStatusRequest`

Output PID status request

**struct**

outputs [list of UUID](#)

#### 26.3.4 `OutputPidStatusResponse`

Response of an output PID status request

**struct**

data

map from **UUID** to **OutputTransportStreamStatus**

### 26.3.5 OutputTransportStreamStatus

Status for the output TS

#### struct

services

map from **int** to **ServiceSourceStatus**

Status for each service in the TS. Map from service ID to status struct.

pids

list of **PidStatus**

Info on all PIDs, both signalled and unsignalled. Map from service PID value to status struct.

### 26.3.6 PidStatus

PidStatus state of PID

#### struct

serviceId

list of **int**

List of serviceIds a PID belong to.

componentType

**optional** **component.ComponentType**

ComponentType of PID

ccErrors

**int**

CC errors on PID

dataRate

**bigint**

Data rate of PID

scrambling

**Scrambling**

Scrambling of PID (none, even or odd)

PCR

**bool**

present

**bool**

True if signalled pid has rate

language

**optional string**

Language of PID, if applicable (e.g. audio pid)

casId

**optional int**

Id of CAS system, if applicable (e.g. ECM pid)

pidIn

**int**

PID number on input

pidOut

**optional int**

PID number on output

### 26.3.7 Scrambling

Scrambling state of PID

#### enum

NONE

No scrambling

EVEN

Scrambling with even control word

ODD

Scrambling with odd control word

### 26.3.8 ServiceSourceStatus

Status for an input service

#### struct

|             |                                                       |
|-------------|-------------------------------------------------------|
| serviceName | <b>optional string</b><br>Service name                |
| pmtPid      | <b>int</b><br>PMT pid for service                     |
| pcrPid      | <b>optional int</b><br>PCR pid for service if present |

### 26.3.9 TransportStreamStatus

Status for the TS

#### struct

|                   |                                                                                                                                                  |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| ccErrors          | <b>int</b><br>Total CC error count for this TS.                                                                                                  |
| services          | <b>map</b> from <b>int</b> to <b>ServiceSourceStatus</b><br>Status for each service in the TS. Map from service ID to status struct.             |
| pids              | <b>map</b> from <b>int</b> to <b>PidStatus</b><br>Info on all PIDs, both signalled and unsignalled. Map from service PID value to status struct. |
| tsId              | <b>optional int</b><br>Transport Stream ID; set if PAT pid present in stream                                                                     |
| originalNetworkId | <b>optional int</b><br>Original Network ID; set if SDT pid present in stream                                                                     |